

New lower bound of the r -th order nonlinearity via algebraic immunity

The the r -th order nonlinearity of Boolean function f over F_2^n is $nl_r(f) = \min_{\deg(g) \leq r} d(f, g)$, where $d(f, g)$ is the Hamming distance. If we look at the Boolean function as a set of its values, the nonlinearity corresponds to the distance to $RM(r, n)$ Reed-Muller codes of order r .

In the case $r > 1$, the known algorithms for computing $nl_r(f)$ work for a foreseeable time only for functions with a very small number of variables [1]. Therefore, there is a request to obtain lower bounds of this quantity via some other properties of Boolean functions. In [2-7] there were obtained the lower bounds of $nl_r(f)$ through the values of algebraic immunity $AI(f) = \min_{f \cdot g \equiv 0 \text{ or } (f+1) \cdot g \equiv 0} \deg(g)$. The lower bounds for the case $r = 1$ and $r = 2$ were exact. The best known lower bound of the r -th order nonlinearity for $r > 2$ via algebraic immunity was obtained in [6]. Now we improve this bound.

Theorem 1. Let f is a Boolean function over F_2^n . Then

$$nl_r(f) \geq \sum_{k=0}^{AI(f)-r-1} \binom{n}{k} + \sum_{k=AI(f)-2r}^{AI(f)-r-1} \binom{n-r}{k} + 2 \sum_{k=AI(f)-2r-1}^{AI(f)-r-2} \binom{n-r-2}{k} + 4 \sum_{k=AI(f)-2r-2}^{AI(f)-r-3} \binom{n-r-4}{k}.$$

Proof. We will use a Theorem from [3]:

$$nl_r(f) \geq \min_{\deg(g) \leq r} \dim B_{AI(f)-1}(g),$$

where $B_m(g) = \{h \mid \deg(h) \leq m, \deg(h \cdot g) \leq m\}$.

Let $\deg(g) = r_0 \leq r$.

Without loss of generality we can consider case then there is monomial $x_1 \cdot x_2 \cdot \dots \cdot x_{r_0}$ in polynomial of g .

Represent g :

$$g(x_1, \dots, x_n) = (x_1 + l_1(x_{r_0+1}, \dots, x_n)) \cdot (x_2 + l_2(x_{r_0+1}, \dots, x_n)) \cdot \dots \cdot (x_{r_0} + l_{r_0}(x_{r_0+1}, \dots, x_n)) + g_1(x_1, \dots, x_n),$$

where all monomials of polynomial g_1 have $r_0 - 2$ or less of variables $x_1, \dots, x_{r_0}$ and $l_1, \dots, l_{r_0}$ are linear functions.

If g_1 is constant then $\dim B_m(g) = \sum_{k=0}^m \binom{n}{k} - \sum_{k=m-r_0+1}^m \binom{n-r_0}{k}$.

If g_1 is not constant choose for it among the monomials of maximum length which contain the maximum number of variables $x_1, \dots, x_{r_0}$ the one that comes first in lexicographic order.

Without loss of generality we can consider case then this monomial is $x_1 \cdot x_2 \cdot \dots \cdot x_{m_1} \cdot x_{r_0+1} \cdot \dots \cdot x_{r_0+r_1-m_1}$, where $m_1 \leq r_0 - 2$ and $r_1 \leq r_0 \leq r$.

Represent g_1 :

$$g_1(x_1, \dots, x_n) = (x_1 + l_1^1) \cdot (x_2 + l_2^1) \cdot \dots \cdot (x_{m_1} + l_{m_1}^1) \cdot (x_{r_0+1} + l_{r_0+1}^1) \cdot \dots \cdot (x_{r_0+r_1-m_1} + l_{r_0+r_1-m_1}^1) + \\ + g_2(x_1, \dots, x_n),$$

where all monomials of polynomial g_2 have $r_0 - 2$ or less of variables $x_1, \dots, x_{r_0}$ and $r_1 - 2$ or less of variables $x_1, \dots, x_{m_1}, x_{r_0+1}, \dots, x_{r_0+r_1-m_1}$. And where l_i^1 are linear functions over variables $x_{m_1+1}, \dots, x_{r_0}, x_{r_0+r_1-m_1}, \dots, x_n$.

If g_2 is constant then $\dim(B_m(g)) \geq \sum_{k=0}^m \binom{n}{k} - \sum_{k=m-r_0+1}^m \binom{n-r_0}{k} - \sum_{k=m-r_1+1}^m \binom{n-r_1}{k}$.

If g_2 is not constant choose for it among the monomials of maximum length $r_2 \leq r_1 \leq r_0$ which with maximum number of variables $x_1, \dots, x_{m_1}, x_{r_0+1}, \dots, x_{r_0+r_1-m_1}$ the one that comes first in lexicographic order.

Without loss of generality we can consider case then this monomial is $x_1 \cdot x_2 \cdot \dots \cdot x_{q_1} \cdot x_{m_1+1} \cdot \dots \cdot x_{m_1+q_2} \cdot x_{r_0+1} \cdot \dots \cdot x_{r_0+q_3} \cdot \dots \cdot x_{r_0+r_1-m_1+r_2-q_1-q_2-q_3}$, where $q_1 + q_3 \leq r_0 - 2$, $q_1 + q_3 \leq r_1 - 2$.

Denote A_1 the set of monomials of $x_1, \dots, x_n$ length less or equal $m - r_0$.

Denote A_2 the set of products $g(x_1, \dots, x_n)$ and monomials of $x_{r_0}, \dots, x_n$ length less or equal $m - r_0$ and more or equal $m - 2r_0$.

Denote A_3 the set of all products $g(x_1, \dots, x_n)$, $1 \leq p \leq r_0 - m_1 - 1$ from linear functions $(x_{m_1+1} + l_{m_1+1} + 1), \dots, (x_{r_0} + l_{r_0} + 1)$ and monomials of $x_{r_0+r_1-m_1}, \dots, x_n$ length less or equal $m - p - r_0$ and more or equal $m - p - 2r_0$.

Denote A_4 the set of all products $g(x_1, \dots, x_n)$, $1 \leq p_1 \leq r_0 - q_1 - q_2$ from linear functions $(x_{q_1+1} + l_{q_1+1} + 1), \dots, (x_{m_1} + l_{m_1} + 1), (x_{m_1+q_2+1} + l_{m_1+q_2+1} + 1), \dots, (x_{r_0} + l_{r_0} + 1)$, $1 \leq p_2 \leq r_0 - q_1 - q_2$ from linear functions $(x_{q_1+1} + l_{q_1+1}^1 + 1), \dots, (x_{m_1} + l_{m_1}^1 + 1), (x_{r_0+1} + l_{r_0+1}^1 + 1), \dots, (x_{r_0+q_3} + l_{r_0+q_3}^1 + 1)$ and monomials of $x_{r_0+r_1-m_1+r_2-q_1-q_2-q_3+1}, \dots, x_n$ length less or equal $m - p_1 - p_2 - r_0$ and more or equal $m - p_1 - p_2 - 2r_0$. Important: for all linear functions in one product, the first variables in parentheses must be different.

$\langle A_1, A_2, A_3, A_4 \rangle \in B_m(g)$ thus

$$\dim \langle A_1, A_2, A_3, A_4 \rangle \leq \dim B_m(g).$$

All functions in $\{A_1, A_2, A_3, A_4\}$ are linearly independent (This is proven from the construction of these sets). Thus

$$\dim B_m(g) \geq |A_1| + |A_2| + |A_3| + |A_4|.$$

The minimum of this amount is achieved if $r = r_0 = r_1 = r_2$, $m_1 = r - 2$, $q_1 + q_3 = r - 2$ and gives lower bound from Theorem 1 for $m = AI(f) - 1$.

References

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